

BACHELOR'S DEGREE EXAMINATION, OCTOBER 2025
2024 ADMISSIONS REGULAR
SEMESTER III - MAJOR (COMPUTER APPLICATIONS)
CA3DSCA02B24 - Data Structures using C

Maximum Marks : 50

44

Time : 1.5 Hours

Part A

(1x10=10)

Answer all questions. Each question carries 1 mark.

1. Explain the concept of a sparse matrix. [CO1, Evaluate]
2. State the purpose of worst-case analysis. [CO1, Analyse]
3. Explain the role of time complexity in algorithm analysis. [CO1, Evaluate]
4. Explain the concept of a linear stack. [CO2, Understand]
5. List one application of a stack in computer science. [CO2, Understand]
6. Explain how a linear stack is implemented using arrays. [CO2, Apply]
7. What is the role of the "head" pointer in a singly linked list? [CO3, Understand]
8. How is a node's data accessed in a linked list? [CO3, Understand]
9. Identify one operation performed on a doubly linked list. [CO3, Remember]
10. Describe a binary tree. [CO4, Remember]

Part B

(5x2=10)

Answer any 2 questions from the bunch of CO1. Each question carries 5 mark.

11. Describe the different types of data structures. Explain the characteristics, advantages, and disadvantages of each. [CO1, Understand]
12. Define time and space complexity of algorithms. Discuss their importance in algorithm analysis with examples of how they are calculated. [CO1, Understand]
13. Explain the concepts of best, average, and worst-case analysis. Provide examples of how they are calculated for different algorithms. [CO1, Understand]

Part B

(5x2=10)

Answer any 2 questions from the bunch of CO2. Each question carries 5 mark.

14. Discuss the infix to prefix expression conversion using a stack. Explain the algorithm step by step and illustrate with an example. [CO2, Apply]
15. Describe the implementation of a queue using arrays. Discuss how enqueue and dequeue operations are performed and the challenges involved in managing the front and rear pointers. [CO2, Understand]
16. Explain the implementation of a circular queue using arrays. Discuss how it resolves the limitations of a regular queue and describe the operations with examples. [CO2, Understand]

Part B

(5x2=10)

Answer any 2 questions from the bunch of CO3. Each question carries 5 mark.

17. Describe the structure and operations of a doubly linked list. How is it different from a singly linked list? [CO3, Apply]
18. Explain how a linked list can be arranged in ascending or descending order of its data. Discuss the algorithms used for sorting linked lists. [CO3, Analyse]
19. Explain the concept of a singly linked list and discuss the insertion and deletion operations in detail. Provide examples. [CO3, Understand]